

LYNSIE WARR

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📍 Orange County, CA

🌐 linsie-warr

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EXPERIENCE

Graduate Student Researcher

University of California, Irvine

📅 September 2021–Present 📍 Irvine, CA

- Developing statistical methodology to model the impacts of climate variables on American agriculture
- Developed simulation study to evaluate spatial point process model of built environmental factors and their effects on public health

Graduate Student Intern

NASA Jet Propulsion Laboratory

📅 June 2024–April 2025 📍 Pasadena, CA

- Modeled effectiveness of Ocean Alkalinity Enhancement (OAE) for marine carbon dioxide removal using Python
- Assessed uncertainty bounds for a reduced-complexity OAE model utilizing multiple data streams for various model inputs
- Manipulated large climate data products as netCDF files in Python

Statistics Teaching Assistant / Grader

University of California, Irvine

📅 September 2022–June 2025 📍 Irvine, CA

- Taught students in large and small group settings
- Assisted student learning by providing feedback on coursework

Graduate Student Researcher

Lawrence Livermore National Laboratory

📅 July 2022–September 2022 📍 Livermore, CA (Remote)

- Modeled spatial distribution of air particles
- Utilized dimension-reduction methods to efficiently model high-dimensional spatial data
- Implemented various machine learning algorithms in R and Python

Research Assistant

Brigham Young University: Statistics Department

📅 September 2017–August 2021 📍 Provo, UT

- Implemented various Bayesian statistical models using MCMC
- Utilized machine learning methods (Random Forest and Neural Nets) for prediction and interpretation
- Manipulated complex remote sensing data sets using R

Research Assistant

Brigham Young University: Enrollment Services

📅 September 2017–August 2019 📍 Provo, UT

- Analyzed and cleaned large datasets for administrative departments
- Presented model output to non-technical audiences and advised improvements for university procedures
- Designed and distributed surveys (using Qualtrics, etc.)
- Automated workflows for data collection, cleaning, and visualization

EDUCATION

PhD Candidate in Statistics (3.98/4.0)

University of California, Irvine

📅 2021 – Present 📍 Irvine, CA

MS Statistics (3.95/4.0)

Brigham Young University

📅 2019 – 2021 📍 Provo, UT

BS Geology & Statistics (3.95/4.0)

Brigham Young University

📅 2014 – 2019 📍 Provo, UT

- Magna Cum Laude

SKILLS

Computing/Software:

R $\LaTeX$ R Markdown ggplot2
Tableau git/GitHub SQL SAS
Python keras TensorFlow C++
MatLab ArcGIS

Research Experience:

Bayesian Statistics Spatial Statistics
Markov Chain Monte Carlo
Machine Learning Data Visualization
Spatial Point Processes
Functional Data Analysis

Coursework:

Machine Learning Bayesian Statistics
Spatial Statistics Statistical Computing
Correlated Data Probability Theory
Advanced Statistical Inference
Generalized Linear Models
Multivariate Statistical Methods
Linear Model Theory Linear Regression
Infectious Disease Models
Stochastic Processes
Mixed Models Analysis of Variance
Survival Analysis Statistical Consulting
Advanced Geology Coursework

SELECTED PROJECTS

Predicting Binding Sites on Proteins with Machine Learning

Paper published in *Journal of Molecular Biology*

- Collaborated with cancer researchers from BYU's chemistry department to develop novel methods to identify binding sites on 14-3-3 proteins
- Implemented a Random Forest algorithm to predict binding site locations
- Developed a web app for use by the cancer research community

Distributional Validation of Precipitation Data Products

Paper published in *Journal of Agricultural, Biological and Environmental Statistics*, multiple invited presentations

- Worked under NASA grant to develop methodology
- Developed Bayesian Spatially Varying Mixture Model and efficient algorithm for validating climate data products in user-defined spatial regions

Early Alert: Predicting Student Success with Machine Learning

- Created complex dashboards using Tableau to aid in student advisement
- Assisted with developing data collection procedures for student lab data

PUBLICATIONS AND PRESENTATIONS

Papers in Progress

- Warr, Lysie R, Veronica J Berrocal, Angela Rigden, and Donatello Telesca (projected 2025). "Bayesian Functional Kernel Regression for Crop Yield Modeling". Target journal: *Annals of Applied Statistics*.
- Warr, Lysie R, Kay Suselj, Dustin Carrol, Daniel Whitt, and Sarah Smith (projected 2025). "Developing a novel model-based uncertainty quantification framework for marine carbon dioxide removal (mCDR) approaches".

Journal Articles

- Johnson, Jacob A, Matthew J Heaton, William F Christensen, Lysie R Warr, and Summer B Rupper (2024). "Fusing Climate Data Products Using a Spatially Varying Autoencoder". In: *Journal of Agricultural, Biological and Environmental Statistics*, pp. 1–14.
- Egbert, CM, LR Warr, KL Pennington, MM Thornton, AJ Vaughan, SW Ashworth, MJ Heaton, N English, MP Torres, and JL Andersen (2023). "The integration of proteome-wide PTM data with protein structural and sequence features identifies phosphorylations that mediate 14-3-3 interactions". In: *Journal of Molecular Biology* 435.2, p. 167890.
- Warr, Lysie R, Matthew J Heaton, William F Christensen, Philip A White, and Summer B Rupper (2023). "Distributional Validation of Precipitation Data Products with Spatially Varying Mixture Models". In: *Journal of Agricultural, Biological and Environmental Statistics* 28.1, pp. 99–116.

Invited Presentations

- Warr, Lysie R, Matthew J Heaton, William F Christensen, Philip A White, and Summer B Rupper (2022). "Distributional Validation of Precipitation Data Products with Spatially Varying Mixture Models". Joint Statistical Meeting: Washington D.C.
- Warr, Lysie R, Matthew J Heaton, William F Christensen, Philip A White, and Summer B Rupper (2022). "Distributional Validation of Precipitation Data Products with Spatially Varying Mixture Models". 5th International Conference on Econometrics and Statistics: Kyoto, Japan.
- Warr, Lysie R, Matthew J Heaton, William F Christensen, Philip A White, and Summer B Rupper (2021). "Distributional Validation of Precipitation Data Products with Spatially Varying Mixture Models". Computational and Methodological Statistics: London, U.K.

Contributed Presentations

- Warr, Lysie R (2024). "Modeling Crop Yield with Climate Variables and Their Interaction". Joint Statistical Meeting: Portland, Oregon.

Posters

- Warr, Lysie R, Matthew J Heaton, William F Christensen, Philip A White, and Summer B Rupper (2022). "Distributional Validation of Precipitation Data Products with Spatially Varying Mixture Models". ENVR 2022 Workshop: Provo, UT.

RECOGNITION AND AWARDS

- 2024 recipient of the Donald Bren School of Information and Computer Sciences' Innovation Fellowship
- 2022 Honorable Mention, ASA: Section on Statistics and the Environment (ENVR) Student Paper Competition
- 2021 BYU Student Research Competition Session Winner
- 2020 BYU Student Research Competition Session Winner
- BYU Geology Department Outstanding Senior Award
- 2019 BYU Student Research Competition Session Winner
- BYU Geology Department Julie M. Baer scholarship
- BYU Heritage Scholarship (Undergraduate 4 year full tuition for academic merit)